

Neuromorphic ASR

Transducer Models on Memristor Hardware

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Outline

Related Works

Motivation

Prerequisites

Experiments and Results

Future Work

Related Works

- ▶ **Sequence Transduction with Recurrent Neural Networks** [Graves 12]
 - ▷ Introduces the Recurrent Neural Network - Transducer (RNN-T, or simply Transducer) architecture, widely used for Automatic Speech Recognition (ASR).
- ▶ **Conformer: Convolution-Augmented Transformer for Speech Recognition** [Gulati & Qin⁺ 20]
 - ▷ Introduces the Conformer encoder, a common choice for ASR encoders.
- ▶ **Synaptogen: A Cross-Domain Generative Device Model for Large-Scale Neuromorphic Circuit Design** [Hennen & Brackmann⁺ 24]
 - ▷ Models measured device behavior for scalable simulation of memristive neural hardware.
- ▶ **Running Conventional Automatic Speech Recognition on Memristor Hardware: A Simulated Approach** [Rossenbach & Hilmes⁺ 25]
 - ▷ Evaluates quantized Connectionist Temporal Classification (CTC)-based ASR under simulated memristor hardware

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- ▶ Memristor hardware enables fast, energy-efficient neural network inference, but introduce analog inaccuracies.
- ▶ CTC-based Conformers have been evaluated on simulated memristor hardware [Rossenbach & Hilmes⁺ 25].
- ▶ Transducers add an autoregressive prediction network and a joint network, which are repeatedly evaluated during inference.
- ▶ **Objective:** characterize how quantization and simulated device noise affect transducer ASR and assess training and decoding countermeasures.

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Overview [Rossenbach & Hilmes⁺ 25]

- ▶ A **memristor** is an analog computing device with a programmable resistance state [Strukov & Snider⁺ 08].
- ▶ Enables fast, energy-efficient implementation of Vector-Matrix Multiplication (VMM) [Wan & Kubendran⁺ 22] —
→ and thus neural networks.
- ▶ Limitations:
 - ▷ Introduces analog inaccuracies [Wouters & Chen⁺ 16].
 - ▷ Production is still in early stages [Infineon Technologies AG 22].
 - ▷ Current experiments are simulated using SynptogenML [Hennen & Brackmann⁺ 24] [Rossenbach & Hilmes⁺ 25].

Resistor:



Memristor:

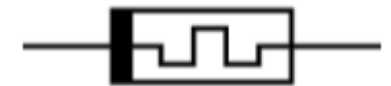


Figure: Resistor and memristor schematics

Analog In-Memory Computing [Aguirre & Sebastian⁺ 24]

- ▶ Multiplication computation done via Ohm's law:

$$V = IR \quad (1)$$

- ▶ Voltage V , current I , and resistance R .
- ▶ Possible to rewrite resistance as conductance G :

$$I = GV \quad \text{with} \quad G = \frac{1}{R} \quad (2)$$

- ▶ Addition done via Kirchhoff's law for any connected node in a circuit, with N connected lines:

$$\sum_{n=0}^N I_n = 0 \rightarrow I_0 = - \sum_{n=1}^N I_n \quad (3)$$

- ▶ Currents can be trivially added by simply connecting electric lines.

Analog In-Memory Computing [Rossenbach & Hilmes⁺ 25]

- ▶ Resistors ($\square$) as connecting element between vertical and horizontal circuit lines
- ▶ Each resistor's conductance is an element of a constant matrix
- ▶ Column sum simply via line connection

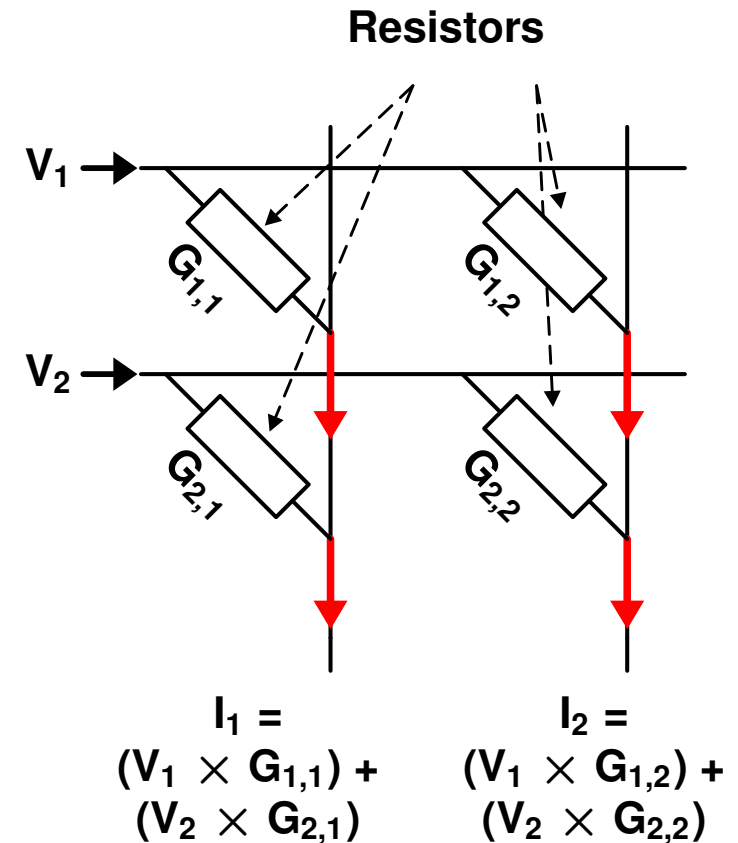


Figure: Crossbar array performing analog VMM [Rossenbach & Hilmes⁺ 25]

The Memristor [Hennen & Brackmann⁺ 24]

Key Idea: Encode each weight bit as a high- or low-resistance state.

- ▶ Let V_S be the set voltage and V_R the reset voltage thresholds.
- ▶ Apply $-V > -V_S \rightarrow$ set to low resistance.
- ▶ Apply $V > V_R \rightarrow$ set to high resistance.
- ▶ Intermediate states possible, but difficult to program reliably.
- ▶ State is physically stored (non-volatile memory).

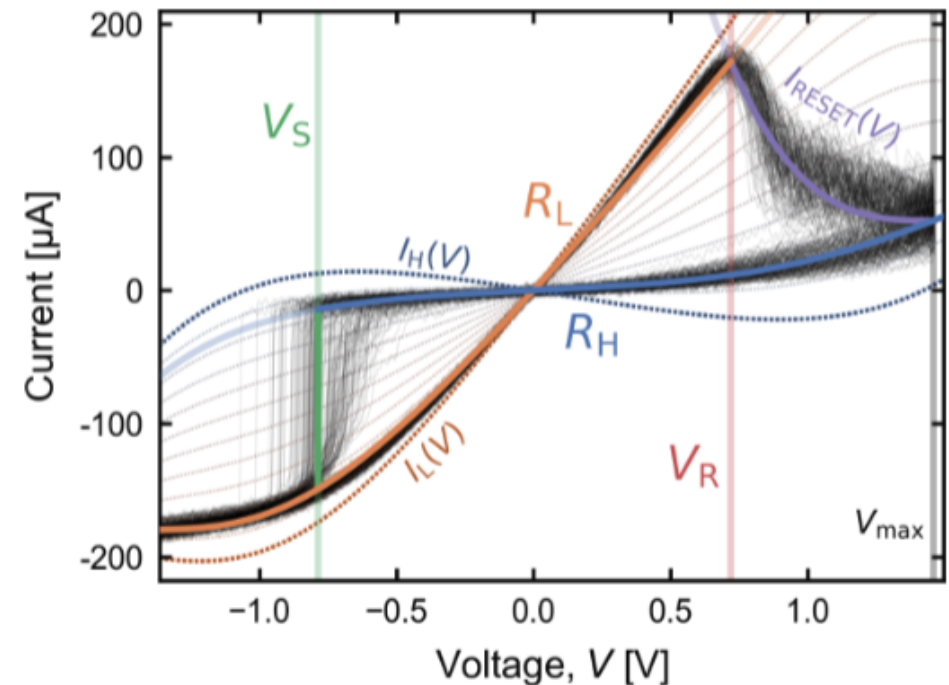


Figure: I–V characteristic curve of a memristor [Hennen & Brackmann⁺ 24]

R_L/R_H : low-/high-resistance state; $I_L(V)/I_H(V)$: corresponding current; V_{max} : maximum applied voltage; $RESET(V)$: current during the reset transition.

Limitations [Rossenbach & Hilmes⁺ 25]

Programming Limitations

- ▶ Resulting resistance state varies with each programming cycle.
- ▶ Non-linear relation of current and voltage.

Cannot exactly model weights

- ▶ Only positive, non-zero weights possible.
- ▶ Current leakage prohibits a true zero.

(Paired) Memristor Array [Aguirre & Sebastian⁺ 24]

- ▶ Use positive and negative bitlines to create memristor pairs
 - ▶ Each row-wise pair of memristors represents a weight
- For N input rows, with row index i and output-column index j :

$$I_j = \sum_{i=1}^N V_i G_{i,j}^+ - \sum_{i=1}^N V_i G_{i,j}^-$$

Problems:

- ▶ Cannot set paired memristors to identical conductances
- Difference close to zero, but not guaranteed

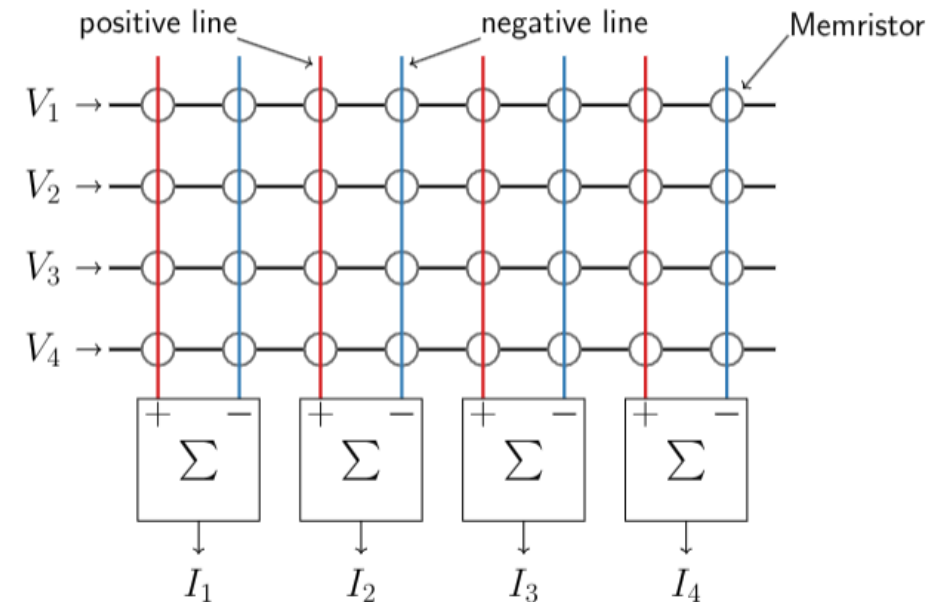


Figure: Paired memristor array structure [Aguirre & Sebastian⁺ 24]

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Transducer Deployment Scenarios

- ▶ Which components of the model should be deployed on the device?
 - ▷ Encoder only vs. encoder + predictor vs. encoder + predictor + joiner (full model).
- ▶ Tradeoff: more offloading $\Rightarrow$ more energy efficiency, but also more noise.

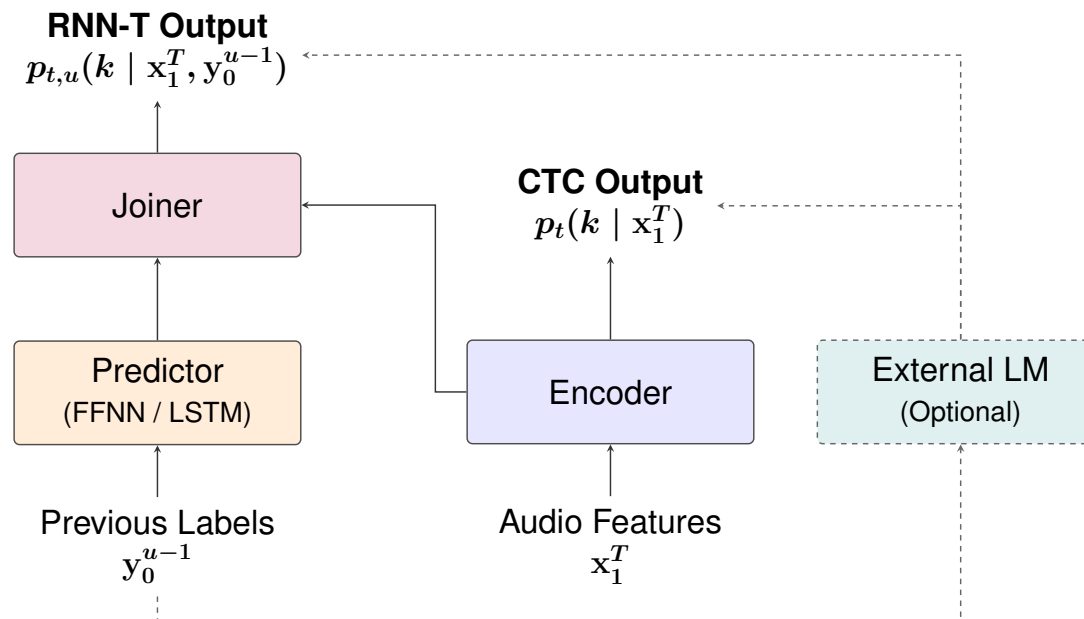


Figure: CTC vs. Transducer architecture components

FFNN: feed-forward neural network; **LSTM**: long short-term memory; **LM**: language model.

k : candidate label or blank; t : acoustic-frame index; u : output-label index.

$x_1^T = (x_1, \dots, x_T)$: audio-feature sequence; $y_0^{u-1} = (y_0, \dots, y_{u-1})$: previously emitted labels;

Experimental Setup

- ▶ **Data**: LibriSpeech [Panayotov & Chen⁺ 15] (960 h), Byte-Pair Encoding (BPE) subword targets (128 merges → 184 labels).
- ▶ **Features**: 80 log-mel filters, 10 ms frame shift; SpecAugment during training.
- ▶ **Encoder**: 12-layer Conformer [Gulati & Qin⁺ 20], 512-dim.
- ▶ Model types:
 1. **Connectionist Temporal Classification (CTC) Model** [Graves & Fernández⁺ 06]: encoder only.
 2. **Feed Forward Neural Network (FFNN)-transducer** [Graves 12]: prediction network is a 2-layer feed-forward net (640-dim) over the previous-label embedding; monotonic alignments [Tripathi & Lu⁺ 19].
 3. **Full-context transducer** [Graves 12]: LSTM prediction network; monotonic alignments [Tripathi & Lu⁺ 19].
- ▶ **Quantization**: Quantization-Aware Training (QAT), per-tensor symmetric, 8- and 4-bit weights, 8-bit activations; zero-point 0.
- ▶ **Stages**: (1) Full-precision baseline → (2) Full pre-training with QAT (fake quantization) → (3) memristor simulation (3–5 runs).

CTC Results

Table: WER (%; mean $\pm$ standard deviation) on LibriSpeech dev-other – CTC. Memristor results use 5 simulation runs.

(a) Baseline

Model	Lex-Free	Tree	
	No LM	No LM	4-gram
CTC	7.1	6.5	5.2

(b) QAT

Bits	Lex-Free	Tree	
	No LM	No LM	4-gram
8	7.3	6.8	5.3
4	8.3	7.6	5.7

(c) Memristor Simulation

Bits	Lex-Free	Tree	
	No LM	No LM	4-gram
8	13.5 $\pm$ 0.37	10.4 $\pm$ 0.20	6.7 $\pm$ 0.09
4	11.9 $\pm$ 0.18	10.1 $\pm$ 0.12	6.8 $\pm$ 0.12

FFNN Transducer Results: 8-bit Quantization

Table: WER (%; mean $\pm$ standard deviation) on LibriSpeech dev-other (8-bit Quantization vs. baseline). Memristor results use 3 simulation runs.

Modules Quantized			Lex-Free	Tree	
Encoder	Predictor	Joiner	No LM	No LM	4-gram
Baseline					
—	—	—	5.9	5.8	4.8
QAT					
✓	—	—	6.3	6.1	4.9
✓	✓	—	6.4	6.2	5.0
✓	✓	✓	6.4	6.2	4.9
Memristor Simulation					
✓	—	—	9.8 $\pm$ 0.06	8.7 $\pm$ 0.03	6.7 $\pm$ 0.07
✓	✓	—	10.7 $\pm$ 0.10	8.9 $\pm$ 0.11	6.7 $\pm$ 0.04
✓	✓	✓	44.7 $\pm$ 1.95	56.1 $\pm$ 10.90	#18.0 $\pm$ 1.27

Result only for 2 runs, other runs WIP.

FFNN Transducer Results: 4-bit Quantization

Table: WER (%; mean $\pm$ standard deviation) on LibriSpeech dev-other (4-bit Quantization vs. baseline). Memristor results use 3 simulation runs.

Modules Quantized			Lex-Free	Tree	
Encoder	Predictor	Joiner	No LM	No LM	4-gram
Baseline					
—	—	—	5.9	5.8	4.8
QAT					
✓	—	—	7.2	6.8	5.2
✓	✓	—	7.3	6.9	5.3 ^{**}
✓	✓	✓	7.3 ^{**}	6.9 ^{**}	5.5 ^{**}
Memristor Simulation					
✓	—	—	10.1 $\pm$ 0.05	9.1 $\pm$ 0.11	6.7 $\pm$ 0.07
✓	✓	—	10.4 $\pm$ 0.11	9.3 $\pm$ 0.12	6.7 $\pm$ 0.06
✓	✓	✓	44.7 $\pm$ 1.95 ^{**}	56.1 $\pm$ 10.90 ^{**}	17.2 $\pm$ 1.99 ^{**}

^{**} Filler result.

FFNN Transducer: Fine-Tuning vs. QAT from Scratch

Table: WER (%; mean $\pm$ standard deviation) on LibriSpeech dev-other: Checkpoint fine-tuning vs. full pre-training (PTN). Configuration – FFNN Transducer with Encoder under QAT. Memristor results use 3 simulation runs.

(a) QAT

Training	Bits	Lex-Free	Tree	
		No LM	No LM	4-gram
Fine-tuned	8	6.6	6.3	5.0
Full PTN	8	6.5	6.2	5.1
Fine-tuned	4	7.2	7.0	5.1
Full PTN	4	7.2	6.8	5.2

(b) Memristor Simulation

Training	Bits	Lex-Free	Tree	
		No LM	No LM	4-gram
Fine-tuned	8	8.1 $\pm$ 0.06	7.5 $\pm$ 0.07	5.8 $\pm$ 0.08
Full PTN	8	9.8 $\pm$ 0.06	8.7 $\pm$ 0.03	6.7 $\pm$ 0.07
Fine-tuned	4	13.1 $\pm$ 0.06	11.2 $\pm$ 0.10	7.2 $\pm$ 0.05
Full PTN	4	10.1 $\pm$ 0.05	9.1 $\pm$ 0.11	6.7 $\pm$ 0.07

Full-Context Transducer Results: 8-bit Quantization

Table: WER (%; mean $\pm$ standard deviation) on LibriSpeech dev-other (8-bit Quantization vs. baseline). Memristor results use 3 simulation runs.

Modules Quantized			Lex-Free	Tree	
Encoder	Predictor	Joiner	No LM	No LM	4-gram
Baseline					
–	–	–	5.8	5.6	4.9
QAT					
✓	–	–	5.9	5.7	5.2
✓	✓	–	6.1	5.9	5.2
✓	✓	✓	6.1	5.9	5.2
Memristor Simulation					
✓	–	–	9.0 ± 0.29	8.1 ± 0.16	7.1 ± 0.07
✓	✓	–	WIP	WIP	WIP
✓	✓	✓	WIP	WIP	WIP

Noisier Models Demand Stronger Internal LM Compensation

Table: WER (%; mean $\pm$ standard deviation) on LibriSpeech dev-other (Tree + 4-gram LM search). Memristor results use 3 simulation runs.

(a) Fixed ELM scale = 0.6

	Modules Quantized		ILM Scale				
	Encoder	Predictor	0.1	0.2	0.3	0.4	0.5
Baseline	—	—	4.8	4.8	4.8	5.0	5.1
QAT	✓	—	5.1	5.1	5.1	5.3	5.4
	✓	✓	5.0	5.0	5.1	5.2	5.4
Memristor	✓	—	17.4 $\pm$ 1.60	8.5 $\pm$ 0.34	7.0 $\pm$ 0.06	6.8 $\pm$ 0.08	6.7 $\pm$ 0.10
	✓	✓	18.7 $\pm$ 1.58	8.8 $\pm$ 0.23	7.2 $\pm$ 0.09	6.8 $\pm$ 0.03	6.9 $\pm$ 0.10

(b) Optimal LM Scales

	Modules Quantized		Scales		WER (%)
	Encoder	Predictor	ILM	ELM	
Baseline	—	—	0.1	0.6	4.8
QAT	✓	—	0.2	0.7	5.0
	✓	✓	0.2	0.7	5.0
Memristor	✓	—	0.3	0.5	6.7 $\pm$ 0.07
	✓	✓	0.4	0.5	6.7 $\pm$ 0.04

ILM: Internal Language Model; **ELM:** External Language Model.

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- ▶ **Fine-tuning:**
 - ▷ Fine-tuning from a full-precision checkpoint for other configurations
- ▶ **Error Analysis:**
 - ▷ Substitutions vs. insertions vs. deletions.
 - How can we improve the degradations in the lexicon-free search?
 - ▷ Study Acoustic posterior distortion under noise.
- ▶ **Generalization:**
 - ▷ Evaluation and training on larger/out-of-domain datasets.

Future Work: Search

► Decoder Parameters:

- ▷ Explore penalties for blank emissions.
- ▷ Tune softmax temperature to sharpen or soften the distribution under noise.

► Search Space vs. Model & QAT Level:

- ▷ Large search space $\Rightarrow$ more hypotheses $\Rightarrow$ more energy consumption and higher Real-Time Factor (RTF).
- ▷ Study the impact of restricting the search space on WER and RTF.
 - $\rightarrow$ How many more hypotheses are needed for on-device (simulated) vs. 4-bit vs. 8-bit vs. full-precision runs?
 - $\rightarrow$ Compare hypothesis expansion across CTC and transducer variants.

► Search Paradigms:

- ▷ Explore alternative search algorithms and language model combinations (e.g., lexicon-free with LSTM/transformer-based LM).

Future Work: Robust Training & Countermeasures

- ▶ **Noise-and-Quantization Aware Training:**
 - ▷ Inject simulated crossbar conductance noise directly into training forward pass.
- ▶ **Selective Layer Mapping:**
 - ▷ Deal with noise in noise-sensitive components (e.g., joint network).
 - Mixed-precision quantization
- ▶ **Denoising & Output Compensation:**
 - ▷ Post-processing to correct analog inaccuracies.
 - Train a small neural network to denoise the memristor output.
 - Use a small calibration dataset to learn a linear transformation to correct the output.

Thank you for your attention!

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In-Memory Computing [Rossenbach & Hilmes⁺ 25]

- ▶ Various technologies exist, with varying tradeoffs:
 - ▷ Chip complexity and density
 - ▷ Energy consumption
 - ▷ Speed and reliability
- ▶ This talk focuses on **memristive** hardware, in particular **resistive RAM (ReRAM)**:
 - ▷ Key benefits: low energy consumption, non-volatility

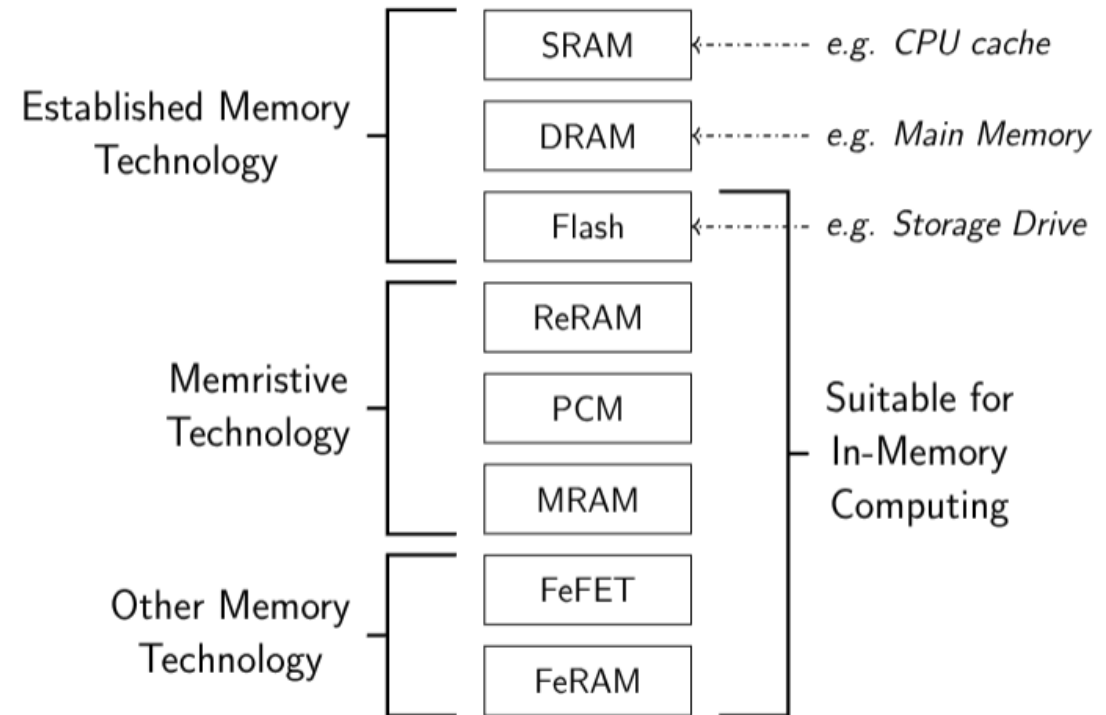


Figure: Taxonomy of In-Memory Computing technologies

Quantization

- ▶ The number of memristors required is proportional to the precision of the weights.
 - ▷ Need to reduce the precision of the weights → quantization.
- ▶ Quantization-Aware Training (QAT) performs better than Post-Training Quantization (PTQ), but requires more training time [Gholami & Kim⁺ 21].
 - High performance gap at low precision

Table: WER (%; mean $\pm$ standard deviation) on TED-LIUMv2 dev, Conformer CTC, 4-gram Language Model (LM) [Rossenbach & Hilmes⁺ 25]

Weight bits	FP32	8	6	5	4	3	2
PTQ	7.2	7.2	7.9	8.9	11.4	30.7	–
QAT		7.3	7.7	7.4	7.8	8.3	22.1

Memristor Neural Layers [Rossenbach & Hilmes⁺ 25]

- ▶ **Tile** the weight matrix:
 - ▷ Tile weight matrix with arbitrary dimensions into smaller fixed-size arrays.
 - ▷ E.g., 512×512 weight matrix $\rightarrow$ 16 crossbar arrays of size 128×128 .
- ▶ **Slice** weights into memristor states:
 - ▷ Binary setting (high vs. low resistance).
 - ▷ Stacking of memristors for each magnitude bit (k -bit weight requires $k - 1$ arrays).
- ▶ **Pair** positive (+) and negative (−) bitlines.

Memristor Neural Layers

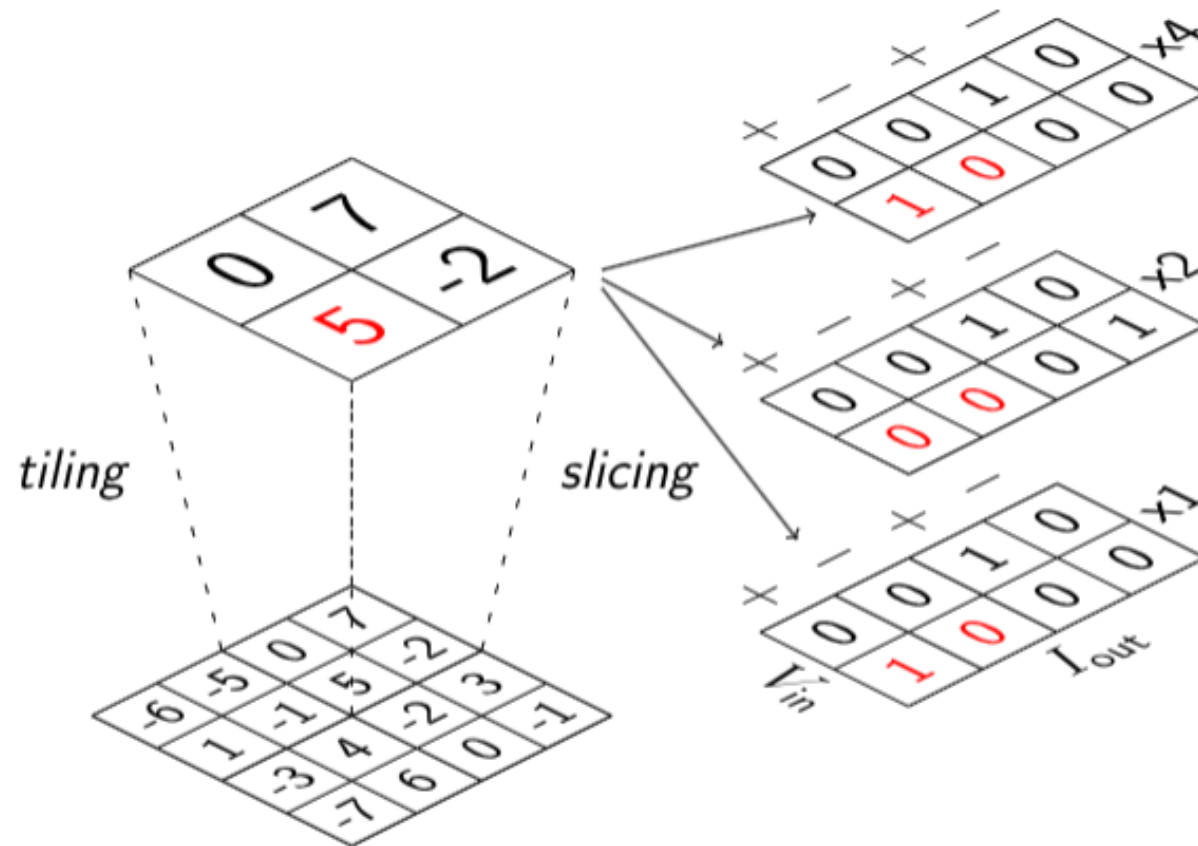


Figure: Mapping a weight matrix on memristor arrays [Rossenbach & Hilmes⁺ 25]

FFNN Transducer: 8-bit vs. 4-bit Quantization

Table: WER (%; mean $\pm$ standard deviation) on LibriSpeech dev-other. Memristor results use 3 simulation runs unless marked otherwise.

Bits	Modules Quantized			Lex-Free	Tree	
	Encoder	Predictor	Joiner	No LM	No LM	4-gram
Baseline						
—	—	—	—	5.9	5.8	4.8
QAT						
8	✓	—	—	6.3	6.1	4.9
8	✓	✓	—	6.4	6.2	5.0
8	✓	✓	✓	6.4	6.2	4.9
4	✓	—	—	7.2	6.8	5.2
4	✓	✓	—	7.3	6.9	5.3 ^{**}
4	✓	✓	✓	7.3 ^{**}	6.9 ^{**}	5.5 ^{**}
Memristor Simulation						
8	✓	—	—	9.8 $\pm$ 0.06	8.7 $\pm$ 0.03	6.7 $\pm$ 0.07
8	✓	✓	—	10.7 $\pm$ 0.10	8.9 $\pm$ 0.11	6.7 $\pm$ 0.04
8	✓	✓	✓	44.7 $\pm$ 1.95	56.1 $\pm$ 10.90	18.0 $\pm$ 1.27 ¹
4	✓	—	—	10.1 $\pm$ 0.05	9.1 $\pm$ 0.11	6.7 $\pm$ 0.07
4	✓	✓	—	10.4 $\pm$ 0.11	9.3 $\pm$ 0.12	6.7 $\pm$ 0.06
4	✓	✓	✓	44.7 $\pm$ 1.95 ^{**}	56.1 $\pm$ 10.90 ^{**}	17.2 $\pm$ 1.99 ^{**}

¹Based on 2 completed runs. ^{**}Filler result.

Full QAT FFNN Transducer Deep Dive

Table: WER (%; mean $\pm$ standard deviation) on LibriSpeech dev-other. The entire model is trained under QAT. Memristor results use 3 simulation runs.

Memristor Simulated Modules			Lex-Free	Tree	
Encoder	Predictor	Joiner	No LM	No LM	4-gram
✓	✓	✓	44.7 ± 1.95	56.1 ± 10.90	18.1 ± 1.27^2
✓	—	✓	45.4 ± 4.99	55.1 ± 6.60	15.0^1
✓	✓	—	9.8 ± 0.03	8.7 ± 0.10	6.7 ± 0.03
✓	✓	1st Layer Only	11.2 ± 0.18	9.4 ± 0.09	6.9 ± 0.07^2

¹Based on 1 completed run; standard deviation unavailable. ²Based on 2 completed runs.

Device VMM Operation Statistics

operation	average	std-dev	min	max
1.0×1	1.016	0.063	0.707	1.27
0.1×1	$0.989 \cdot 10^{-1}$	$0.062 \cdot 10^{-1}$	$0.688 \cdot 10^{-1}$	$1.24 \cdot 10^{-1}$
0.01×1	$0.985 \cdot 10^{-2}$	$0.069 \cdot 10^{-2}$	$0.721 \cdot 10^{-2}$	$1.20 \cdot 10^{-2}$
1.0×0	$8.52 \cdot 10^{-4}$	$4.75 \cdot 10^{-2}$	-0.2751	0.2455
1.0×0.5	0.476	0.094	0.109	1.06

Table: Hardware measurement statistics for simulated memristor VMM operations [Rossenbach & Hilmes⁺ 25].